A Fourier criterion and phase engineering for optimal quantum lifts of abelian Cayley chains

Candidate substantial partial answer to arXiv:2505.12187

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Status. Candidate substantial partial result, likely valid. The open matrix problem is solved for finite abelian Cayley generators; arbitrary symmetric Q-matrices remain open.

Source question

Li and Lu [1] construct a quantum lift of a symmetric finite-state generator Q. On source PDF page 31, after equations (4.21)–(4.22), they ask which families of Q-matrices admit constants

$$M \preceq C_1 Q^2 + C_2(-Q), \qquad \sqrt{C_1} + \sqrt{C_2/\lambda_Q} = O(1), \tag{1}$$

where M is the Gram matrix of $y \mapsto [H,[H,\operatorname{diag} y]]$ and λ_Q is the gap of $-Q$.

$$\Omega\left(\frac{\sqrt{\lambda_Q}}{1+K_1+K_2\lambda_Q^{-1/2}}\right) \le \nu(\mathcal{L}_\gamma) \le \mathcal{O}(\sqrt{\lambda_Q}), \tag{4.18}$$

where λ_Q denotes the spectral gap of Q, and the constants $K_1, K_2 > 0$ satisfy

$$\|[H,[H,Y]]\|_{2,\frac{1}{n}} \le K_1 \|Qy\|_{\ell^2(\nu)} + K_2\sqrt{-\langle y, Qy\rangle_{\ell^2(\nu)}}, \tag{4.19}$$

for $Y = \sum_j y_j |j\rangle\langle j| \in \ker(\mathcal{L}_S)$.

The convergence rate estimate (4.18) follows from arguments analogous to those for (3.25), while the key inequality (4.19) represents a simplified version of (3.32). To obtain optimal constants (K_1, K_2), we reformulate (4.19) in terms of H and Q. We compute:

$$\begin{aligned}[H,[H,Y]] &= \sum_j y_j [H,[H,P_j]] \\ &= \sum_j x_j \left(H^2P_j - 2HP_jH + P_jH^2\right),\end{aligned}$$

for $Y = \sum_j y_j P_j$ with $P_j = |j\rangle\langle j|$, and then

$$\|[H,[H,Y]]\|^2_{2,\frac{1}{n}} = \frac{1}{n}\sum_{i,j} x_i^* x_j M_{ij},$$

where, by noting (h_{ij}) in (4.17) is real symmetric, the positive semidefinite matrix M has entries:

$$M_{ij} = 2\langle i|H^4|j\rangle\delta_{ij} - 8\langle i|H^3|j\rangle h_{ij} + 6\langle i|H^2|j\rangle^2. \tag{4.20}$$

To establish (4.19), it suffices to find positive constants C_1, C_2 such that:

$$M \preceq C_1 Q^2 - C_2 Q, \tag{4.21}$$

which immediately gives the desired inequality with $K_1 = \sqrt{C_1}$ and $K_2 = \sqrt{C_2}$.

It is important to note that verifying (4.21) only requires Q, as the definition of M in (4.20) depends solely on Q through (4.17). To achieve the optimal (quantum) lift with a quadratic speedup for the mixing of $\exp(tQ)$, it is necessary that (4.21) holds with

$$\sqrt{C_1} + \sqrt{C_2/\lambda_Q} = \Theta(1). \tag{4.22}$$

An interesting open question is to characterize the class of Q-matrices for which this property holds. For specific examples with explicit forms of Q, this may be verified through direct computation. For instance, as we show next, in the case of a simple random walk on the chain as in [DHN00], the above construction indeed yields an optimal lift. We consider the simple

Proof intuition

The second term in (1) does not enlarge the asymptotic class: on mean-zero vectors, $\lambda_Q(-Q) \preceq Q^2$. Thus the problem is a uniform "discrete Hessian versus Laplacian" estimate. For an abelian Cayley chain, translation invariance diagonalizes this Hessian Gram matrix. Its Fourier symbol is unusually simple: an averaged fourth power of differences of the Hamiltonian symbol. That identity gives exact constants.

It also exposes a coherent-phase effect. Positive square roots make all complete-graph amplitudes add at the trivial character, producing a constant $|G|/2$. Cubic phases spread the Fourier mass; Weil's exponential-sum bound then makes the same complete-graph lift uniformly optimal. The part still open is a comparably structural characterization without translation invariance.

Exact abelian Fourier criterion

Let G be a finite abelian group, $N = |G|$, and let

$$(Qf)(x) = \sum_{s\in G} q(s)\big(f(x+s) - f(x)\big), \quad q(0) = 0, \quad q(-s) = q(s) \geq 0,$$

be ergodic. Choose $h(0) = 0$ with

$$h(-s) = \overline{h(s)}, \qquad 2|h(s)|^2 = q(s),$$

and let $H_{x,z} = h(z - x)$. Thus $H = H^*$, and the source's overdamped calculation gives exactly Q; its canonical choice is $h(s) = \sqrt{q(s)/2}$. For a character χ_η, put

$$a(\eta) = \sum_{s\in G} h(s)\overline{\chi_\eta(s)}, \qquad \lambda(\xi) = \sum_s q(s)\big(1 - \chi_\xi(s)\big).$$

Symmetry makes both quantities real, and $Q\chi_\xi = -\lambda(\xi)\chi_\xi$. Define M_h by

$$\langle f, M_h f\rangle_{\ell^2(G)} = \|[H,[H,\operatorname{diag} f]]\|^2_{\mathrm{HS},N},$$

using normalized counting and normalized Hilbert–Schmidt norms.

Theorem 1 (Fourier diagonalization)**.** *Every character χ_ξ is an eigenvector of M_h, with eigenvalue*

$$m(\xi) = \frac{1}{N}\sum_{\eta\in\widehat{G}} |a(\eta) - a(\eta-\xi)|^4. \tag{1}$$

Consequently the least C for which $M_h \preceq CQ^2$ on constants perpendicular is

$$\mathcal{C}(Q,h) = \max_{\xi\neq 0} \frac{N^{-1}\sum_\eta |a(\eta) - a(\eta-\xi)|^4}{\lambda(\xi)^2}. \tag{2}$$

For a family of such generators, the source condition (1) holds uniformly if and only if $\sup \mathcal{C}(Q,h) < \infty$.

Proof. Take $f = \chi_\xi$ and write the matrix entry with $z = x + d$ and the intermediate vertex $x + s$. A direct expansion gives

$$[H,[H,\operatorname{diag} f]]_{x,x+d} = \chi_\xi(x) b_\xi(d),$$

where

$$b_\xi(d) = (1 + \chi_\xi(d))(h * h)(d) - 2\big((h\chi_\xi) * h\big)(d).$$

With the displayed convention for a, the Fourier transform is the perfect square

$$\widehat{b}_\xi(\eta) = a(\eta)^2 + a(\eta-\xi)^2 - 2a(\eta)a(\eta-\xi) = (a(\eta) - a(\eta-\xi))^2.$$

Translation covariance makes the characters eigenvectors of the Gram operator. Parseval now gives (1), and simultaneous diagonalization with Q^2 gives (2).

It remains to compare with (1). Put $L = -Q$. On constants perpendicular, $L \succeq \lambda_Q I$, hence $\lambda_Q L \preceq L^2$. If (1) holds and its displayed square-root sum is at most B, then

$$M_h \preceq C_1 L^2 + C_2 L \preceq 2B^2 L^2.$$

Conversely $M_h \preceq CL^2$ gives (1) with $C_1 = C, C_2 = 0$. $\square$

Sharp examples

Corollary 2 (hypercubes)**.** *For the normalized walk on $G = (\mathbb{Z}/2\mathbb{Z})^d$, with $q(e_j) = 1/d$ and the canonical positive h, one has*

$$\mathcal{C}(Q, h) = 3 - \frac{2}{d}.$$

Thus this family satisfies the source's optimal-lift criterion uniformly.

Proof. If ξ has Hamming weight k, then $\lambda(\xi) = 2k/d$ and

$$a(\eta) - a(\eta - \xi) = \sqrt{\frac{2}{d}} \sum_{j \in \operatorname{supp} \xi} (-1)^{\eta_j}.$$

For a sum S_k of k independent Rademacher variables, $\mathbb{E} S_k^4 = 3k^2 - 2k$. Hence

$$\frac{m(\xi)}{\lambda(\xi)^2} = \frac{4(3k^2 - 2k)/d^2}{4k^2/d^2} = 3 - \frac{2}{k}.$$

The maximum occurs at $k = d$. □

Proposition 3 (complete graph: obstruction and repair)**.** *Let G have order N and $q(s) = 1/(N-1)$ for every $s \neq 0$.*

(i) *For the canonical positive choice $h(s) = 1/\sqrt{2(N-1)}$, one has the sharp identity*

$$M_h = \frac{N}{2} Q^2.$$

In particular every pair C_1, C_2 in (1) satisfies $\sqrt{C_1} + \sqrt{C_2/\lambda_Q} \geq \sqrt{N/2}$.

(ii) *If $G = \mathbb{Z}/p\mathbb{Z}$ with prime $p \geq 5$, set, for $s \neq 0$,*

$$h(s) = \frac{\exp(2\pi i s^3/p)}{\sqrt{2(p-1)}}.$$

Then $H = H^*$, it has the same overdamped complete-graph generator Q, and

$$M_h \preceq 256 Q^2.$$

Thus Hamiltonian phases restore a dimension-free optimal-lift estimate.

Proof. For (i), the Hamiltonian symbol equals $\sqrt{(N-1)/2}$ at the trivial character and $-1/\sqrt{2(N-1)}$ elsewhere. For nontrivial ξ, only $\eta = 0, \xi$ contribute to (1). Since $\lambda(\xi) = N/(N-1)$, formula (2) gives exactly $N/2$. All nonconstant characters have the same eigenvalue, proving the operator identity and the lower bound in (1).

For (ii), $h(-s) = \overline{h(s)}$ and $2|h(s)|^2 = 1/(p-1)$. Its symbol is a cubic additive-character sum with the $s = 0$ term removed. Weil's bound for a degree-three polynomial over $\mathbb{F}_p$ [2] yields

$$|a(\eta)| \leq \frac{2\sqrt{p} + 1}{\sqrt{2(p-1)}} < 2.$$

Thus (1) gives $m(\xi) < 16 \cdot 2^4 = 256$. Here $\lambda(\xi) = p/(p-1) > 1$ for $\xi \neq 0$, so $m(\xi)/\lambda(\xi)^2 < 256$, and Theorem 1 finishes the proof. □

Verification, limitations, and novelty

The included script checks the exact complete-graph and hypercube constants and evaluates the cubic construction for primes through 101; all assertions pass. These computations are regression tests, not proof. The argument above uses only Fourier inversion, a Rademacher fourth moment, and the classical cubic Weil bound.

The result characterizes all abelian Cayley instances but not arbitrary symmetric matrices. It also shows that the answer depends on Hamiltonian phases: part (i) concerns exactly the positive square root in source equation (4.17), while part (ii) uses the more general self-adjoint Hamiltonian allowed by the source's preceding overdamped construction.

Consequences for the source problem

First, within the abelian Cayley class the two constants in (4.21) carry no more asymptotic information than the single fourth-difference ratio (2); the C_2 term can always be absorbed into $C_1 Q^2$ at a universal loss once (4.22) is imposed. Thus (2) is an exact and directly computable answer for this class, not merely a sufficient test.

Second, small spectral gap and large degree are neither individual obstructions. Hypercubes have gap $2/d$ and degree d but a constant below 3, whereas the normalized complete graph has gap tending to one but the canonical constant is $N/2$. The relevant datum is coherent concentration of the square-root Hamiltonian symbol.

Third, a characterization purely in terms of Q necessarily presupposes a phase convention such as source equation (4.17). The same complete-graph Q fails for the positive convention and succeeds for cubic phases. This suggests separating two future questions: classify the canonical lift, and optimize over all self-adjoint Hamiltonians with $2|h_{xz}|^2 = Q_{xz}$.

On 11 August 2026, the run registry, solution, attempt, and proof-gap indexes were searched for the arXiv id and core quantum-lift/Fourier terms; no answer was found. No broad external literature search was performed, so novelty confidence is moderate-to-low.

Human-review recommendation. Check the Fourier shift convention in (1), the deletion of the zero term in the Weil estimate, and the application of the source's abstract lift theorem to the phase-modified self-adjoint Hamiltonian. The general non-Cayley problem remains open.

References

[1] B. Li and J. Lu, *Speeding up quantum Markov processes through lifting*, arXiv:2505.12187 (2025), especially Section 4.2 and equations (4.17)–(4.22).

[2] A. Weil, *On some exponential sums*, Proc. Natl. Acad. Sci. USA 34 (1948), 204–207.